

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 9, 2013 2:10-3:40 PM

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In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) A spacecraft is instantaneously located at the midway point between the Earth and Moon and the spacecraft has zero velocity vector with respect to the Earth-Moon barycentric coordinate frame. Based on the non-dimensionalized equations of motion for the restricted three-body dynamics, answer the following questions:

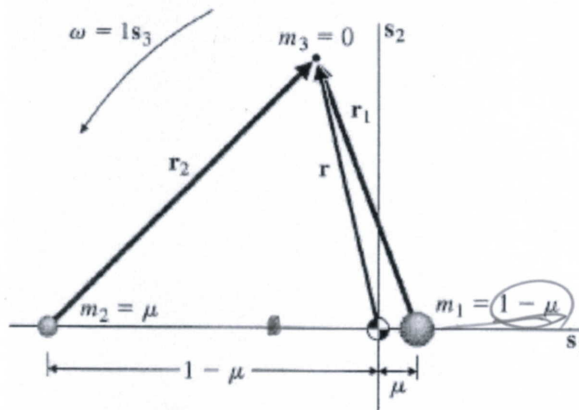


FIGURE 10.1

Geometry of the restricted problem.

$$\ddot{x} - 2\dot{y} - x = -\frac{(1-\mu)(x-\mu)}{r_1^3} - \frac{\mu(x+1-\mu)}{r_2^3}$$

$$\ddot{y} + 2\dot{x} - y = -\frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3}$$

$$r_1 = \sqrt{(x-\mu)^2 + y^2 + z^2}, \quad r_2 = \sqrt{(x+1-\mu)^2 + y^2 + z^2}$$

$$\mu = 0.0123 \text{ for the Earth-Moon System}$$

- A. (15) Is the spacecraft in equilibrium at this midway point between the Earth and Moon? If yes, provide your rationale. If not, what is the acceleration vector of the spacecraft with respect to the barycentric coordinate frame?
- B. (5) What is the constant (control) acceleration vector that is required to keep the spacecraft in equilibrium at this midway point between the Earth and Moon? Express your control acceleration vector in the barycentric coordinate frame.

2. (20) For the 3-2-1 rotation, derive the angular velocity components $\vec{\omega} = [\omega_x \ \omega_y \ \omega_z]^T$ in terms of the Euler angles (ψ, θ, ϕ) and their rates.

3. (30) Euler's Equations of Motion

- A. (15) Starting from Newton's law that the inertial time rate of change of the angular momentum is the total applied torque, derive Euler's equations of motion in a body fixed frame, which is assumed to be principal. The moment of inertia in the given set of principal axes, the angular velocity component and the external moment component are given as $I = \text{diag}(A, B, C)$, $\vec{M} = [M_1 \ M_2 \ M_3]^T$ and $\vec{\omega} = [\omega_1 \ \omega_2 \ \omega_3]^T$, respectively.
- B. (15) In case the rotational motion is torque-free, show that the kinetic energy and the magnitude of angular momentum are conserved.

4. (20) Consider the pitch axis control model for a spacecraft in a circular orbit given by

$$I_y \Delta \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \Delta \theta + M_y$$

where the orbital frequency $\omega_0 = 0.0012 \text{ rad/sec}$, the spacecraft principal moments of inertia are $I = \text{diag}(150, 100, 200) \text{ kg} \cdot \text{m}^2$. In this equation $\Delta \theta$ denotes the pitch angle and M_y denotes the pitch axis control moment.

- A. If the pitch axis control moment $M_y = 0$, show that the pitch equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ is an unstable equilibrium.
- B. Show that linear state feedback to the pitch axis control moment can be used to stabilize the pitch axis equilibrium $(\Delta \theta, \Delta \dot{\theta}) = (0, 0)$ of the spacecraft. Choose feedback gains so that the equilibrium of the closed loop is asymptotically stable with damping ratio of 0.7 and natural frequency of 0.008 rad/sec (that is, the closed loop characteristic polynomial is $s^2 + 0.0112s + 0.000064$).

5. (10) Briefly discuss rotation matrix (direction cosine matrix), Euler angles and quaternions.

$$\begin{aligned} I_x \dot{\omega}_x &= (I_y - I_z) \omega_y \omega_z + M_x \\ I_y \dot{\omega}_y &= (I_z - I_x) \omega_z \omega_x + M_y \\ I_z \dot{\omega}_z &= (I_x - I_y) \omega_x \omega_y + M_z \end{aligned}$$

$$\begin{bmatrix} \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

여기에서 stability는 $\Delta \theta$ 를 얻고 characteristic e.a.를 보아야 하는 것이다.

$$s^2 = \frac{3\omega_0^2 (-I_x + I_z)}{I_y} \text{ 가 되는 것이다.}$$